

Quantum Convolutional Neural Network for Classical Data Classification

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Abstract

Convolutional Neural Networks (CNNs) are the standard for image classification but require substantial computational resources due to large parameter counts. Quantum Convolutional Neural Networks (QCNNs) offer a parameter-efficient alternative suitable for Noisy Intermediate-Scale Quantum (NISQ) devices. This project implements a hybrid QCNN to benchmark classification accuracy and parameter efficiency against classical ANNs, CNNs, and Vision Transformers (ViTs) on standard datasets. Experimental results demonstrate that QCNNs achieve meaningful accuracy (>93%) using over an order of magnitude fewer trainable parameters than their classical counterparts, highlighting their potential for near-term quantum machine learning.

1. Introduction

Quantum computing introduces novel primitives for data representation. QCNNs adapt the inductive biases of classical CNNs—specifically local connectivity and translational invariance—into shallow quantum circuits. While classical CNNs utilize millions of parameters, QCNNs exploit the high-dimensional Hilbert space of qubits to capture complex correlations with minimal weights.

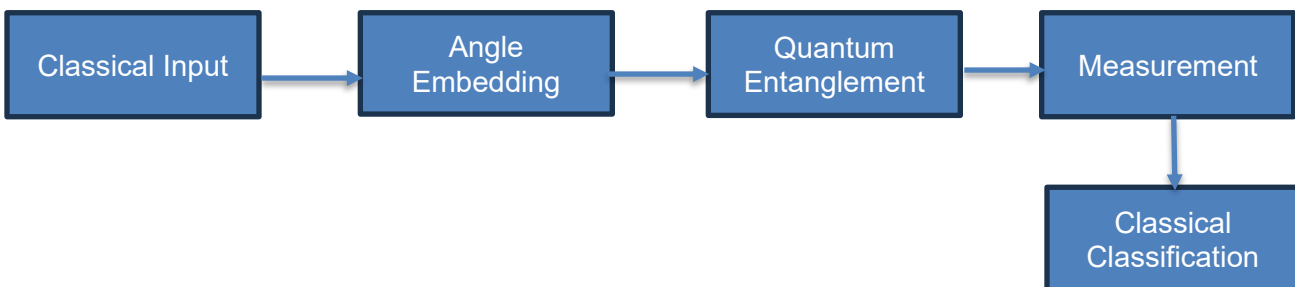
Building on work by Hur et al. [1], who demonstrated fully parameterized QCNNs for MNIST, this project adopts a hybrid quantum-classical approach. We replace the classical convolution operation with a quantum circuit consisting of **Angle Embedding** and **Strongly Entangling Layers**, aiming to analyze the trade-off between expressivity and resource cost.

2. QCNN Architecture

The architecture processes classical image data using a "sliding quantum kernel." The pipeline consists of three main stages:

- **Classical Preprocessing:** The image is divided into small patches (e.g., 2×2 pixels).
- **Quantum Convolution:** Each patch is mapped to a quantum state, processed by a variational circuit, and measured to extract features.
- **Classical Classification:** The resulting feature map is fed into a classical fully connected layer for final prediction.

The system uses $N=4$ qubits to process each 2×2 patch.

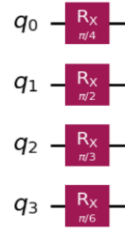


3. Quantum Implementation

3.1 Angle Embedding

Data encoding is critical for quantum processing. We normalize input pixel values to the range $[0, \pi]$ and encode them into qubit rotation angles.

Circuit: We use R_x rotation gates, where each feature x_j determines the rotation angle for qubit j .



Mathematical Derivation:

The objective is to map a classical vector $X=[x_1, x_2, \dots, x_N]$ to a quantum state $|\psi\rangle$.

- **Initial State:** All qubits start in the ground state

$$|0\rangle^{\otimes N} = |0\rangle_1 \otimes |0\rangle_2 \otimes \dots \otimes |0\rangle_N$$

- **Gate Operation:** A rotation $R_x(x_j)$ is applied to each qubit. The matrix form is:

$$R_x(\theta) = \begin{pmatrix} \cos(\frac{\theta}{2}) & -i \sin(\frac{\theta}{2}) \\ -i \sin(\frac{\theta}{2}) & \cos(\frac{\theta}{2}) \end{pmatrix}$$

- **Final State:** The system evolves into a tensor product of rotated states:

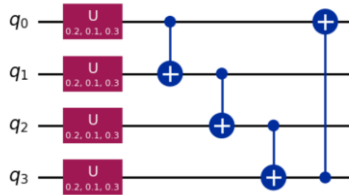
$$|\psi\rangle = \bigotimes_{j=1}^N \left(\cos\left(\frac{x_j}{2}\right) |0\rangle_j - i \sin\left(\frac{x_j}{2}\right) |1\rangle_j \right)$$

This creates a superposition where amplitudes encode the pixel intensity.

3.2 Strongly Entangling Layers

To extract features, we employ a variational ansatz consisting of rotational layers and entangling gates.

Circuit: Each layer consists of general rotations $U(\theta, \phi, \omega)$ on all qubits, followed by a cascade of CNOT gates arranged in a nearest-neighbour topology.



Mathematical Derivation:

For a single layer l acting on N qubits:

- **Rotation Step** (U_{ROT}): Each qubit j undergoes a rotation defined by parameters $\theta_{l,j}$:

$$U_{ROT}^{(l)} = \bigotimes_{j=1}^N R_z(\omega_{l,j}) R_y(\theta_{l,j}) R_z(\phi_{l,j})$$

- **Entanglement Step** (U_{ENT}): CNOT gates connect qubit j (control) to $(j+r) \pmod N$ (target), where r is the range

$$U_{ENT}^{(l)} = \prod_{j=1}^N \text{CNOT}(j, (j+r)\%N)$$

hyperparameter (often increases with layer depth, e.g., $r=2l-1$).

- **Total Evolution:** For L layers, the final unitary is

$$U_{total} = \prod_{l=1}^L (U_{ENT}^{(l)} \cdot U_{ROT}^{(l)})$$

The entangling gates create non-local correlations (e.g., transforming product states into Bell-like states), allowing the model to detect spatial features that single-qubit filters cannot.

4. Experimental Setup

The model was implemented using **PennyLane** for quantum simulation and **PyTorch** for the hybrid training loop.

- **Optimizer:** Adam with Cross-Entropy Loss.
- **Configuration:** N=4 qubits, L=2 layers for the quantum circuit.
- **Data:** Training and testing were conducted on standard digit datasets.
- **Experiments were conducted on:**
 - MNIST: handwritten digit images (28×28 grayscale)
 - Fashion-MNIST: clothing images with higher intra-class variability
 - Due to the computational cost of quantum circuit simulation, QCNN training on Fashion-MNIST was performed using a randomly subsampled set of 5,000 training images.
 - ANN and CNN models were trained on the full datasets. The QCNN was implemented using PennyLane with PyTorch integration, enabling hybrid quantum–classical training. Optimization was performed using the Adam optimizer with cross-entropy loss. Evaluation was conducted on the full test sets.

5. Complexity & Comparison

Parameter Efficiency:

The QCNN demonstrates extreme parameter efficiency.

- **Classical CNN:** A standard 3 x 3 kernel with minimal channels requires ~80+ parameters.
- **QCNN:** With N=4 and L=2, the kernel requires only 3 x N x L = 24 trainable weights.

Computational Complexity:

- **Classical:** $O(H \cdot W \cdot k^2)$, dominated by arithmetic MAC operations.
- **Quantum:** $O(S \cdot L \cdot N)$, where S is the number of "shots" (measurements). While the parameter count is low, the measurement process introduces a constant overhead that impacts training speed compared to highly optimized GPU matrix multiplications.

6. Results (Digits Dataset)

Model	Architecture	Accuracy	F1-Score
Classical ViT	Vision Transformer	96.67%	0.9664
Classical ANN	MLP	96.67%	0.9663
Classical CNN	ConvNet	94.72%	0.9467
Hybrid QCNN	Quant Conv + Linear	93.06%	0.9312
Hybrid Q-ViT	Quant Patch + Transf	86.94%	0.8687
Hybrid QANN	Flatten + VQC	67.50%	0.6242

The **Hybrid QCNN** achieves 93.06% accuracy, comparable to the Classical CNN (94.72%), despite using roughly **44x fewer parameters**. The Q-ViT and QANN variants underperformed, suggesting that the inductive bias of convolution is crucial for image tasks, whether implemented classically or quantumly.

Dataset	Model	Accuracy(%)	Parameters
MNIST	ANN	98	235146
MNIST	CNN	98.81	421642
MNIST	QCNN	82.35	9482
Fashion-MNIST	ANN	88.74	235146
Fashion-MNIST	CNN	92.06	421642
Fashion-MNIST	QCNN	59.35	9482

Accuracy vs Parameter Efficiency Summary

- CNN achieves the highest accuracy but requires more than 420k parameters.
- QCNN achieves meaningful accuracy using only 9,482 parameters.
- QCNN uses approximately 44× fewer parameters than CNN.
- Performance degradation is more pronounced on Fashion-MNIST due to increased data complexity.

7. Conclusion

The results validate that QCNNs can achieve competitive accuracy with a significantly compressed parameter space.

- **Strengths:** The quantum kernel maps data into a 16-dimensional Hilbert space (2^4), potentially capturing complex correlations inaccessible to linear filters.
- **Limitations:** The current approach is limited by the simulation overhead of quantum measurement and the shallow depth of NISQ circuits. It does not yet claim "quantum advantage" in terms of speed.

In conclusion, QCNNs represent a promising direction for resource-constrained environments. While classical networks currently offer slightly higher absolute accuracy, the high expressivity-to-parameter ratio of QCNNs makes them a vital candidate for future hybrid quantum-classical machine learning.

References

[1] T. Hur, L. Kim, and D. K. Park, "Quantum Convolutional Neural Network for Classical Data Classification," *Quantum Machine Intelligence*, 2022.

Appendix:



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